

## **Cryptography**

**Cryptography is the method of protecting information by transforming it into a secure format, a encrypted message in which letters are replaced with other characters.**

**Some terms :**

**Plain text : Unencrypted message**

**Cipher text : Encrypted message**

**Encryption : Process of converting plain text into cipher text.**

**Decryption : Process of converting cipher text into plain text.**

**Algorithm : The method used to encrypt/decrypted the plain text.**

**Key : The data used for encrypting/decrypting.**

**There are some cryptographic algorithms present, generally we categorise them as follows -**

**Symmetric Cryptography :**

**In this one single key is used for encryption and the same key is used for decryption. DES is the example of symmetric key cryptography.**

**Asymmetric cryptography/Public key cryptography :**

**In this two keys are used, Public key is used for encryption and private key is used for decryption like RSA.**

**Substitution cipher :**

**Substitution cipher is a method of encrypting in which units of plaintext are replaced with the ciphertext.**

**Caesar Cipher works by adding or subtracting 3 or any characters. That means if in a Plain Text there is a Character E either it will be transferred it to B and if the character is A it will be transferred to X.**

**This Cipher algorithm is having some mathematical equations which describe the functionality of a cryptography process.**



One more thing you can give alphabets and substraction, addition value according to your need. You can do addition to subtraction and subtraction to addition too to need.

**Example Of Ceaser Cipher :**

**KEY : ABCDEFGHIJKLMNOPQRSTUVWXYZ**

**Encryption Algo : Subtraction and addition of 3 character of number**

**Substraction :**

**Plain text : GOURAV**

**Cipher text : DLROXS**

**Addition :**

**Plain text : DLROXS**

**Cipher text : GOURAV**

**Decryption Algo : Addition and substraction of 3 character of number**

**Addition :**

**Plain text : GOURAV**

**Cipher text : JRXUDY**

**Substraction :**

**Plain text : JRXUDY**

**Cipher text : GOURAV**

**Data Encryption Standard (DES)**

**Introduced in 1975**

**Standardized in 1977**

**Problem with DES: short key length (56 bits)**

Now considered as insecure

Improved version: Triple DES (involves DES three times)

Problem with Triple DES: slow, compute heavy

Advanced Encryption Standard (AES)

First published in 1998

Became a federal government standard in 2002

First approved (and only) publicly accessible cipher approved by the NSA for top secret information.

## Steganography

Steganography is a process in which we basically hide a data inside a data. It is just a process of embedding a secret piece of text within a text, picture, or audio.

Commands for windows:

```
CMD > copy /b sunset.jpg+secret.txt sunsetsecret.jpg
```

Here, /b is used for Binding the 2 files, Copy is used for copying the content of second file to first file.

```
#copy /b sunset.jpg+secret.txt sunsetsecret.jpg
```

Commmand for linux :

```
#steghide embed -cf sunset.jpg -ef secret.txt - For hiding txt file data into this image.
```

Now enter the password you want to give for this file.

```
#steghide extract -sf sunset.jpg - For extracting data from image
```

Enter the password and you will get the txt file.

```
#steghide info kevinmitnick.jpg
```

## Hashes :

A hash is a function that converts one value to another. But there is a difference between a cipher encryption and a hash. The difference is encrypted text can be reverted and further decrypted, but hashes cannot be reverted. We need to crack the hashes.

Hash function is that which takes an input and returns a fixed-size alphanumeric string. The string is called the hash value. Examples MD5 Hash.

EG. alphanumeric - scusege67dg367df7fd3fd37f3636d

## Cracking methods for Hashes :

We have to create a dictionary and have to convert every word into the hash of a particular wordlists, and after that we will compare that particular hash. If matches it means that the specific word is found. Hashes are usually uniques.

## Passwords & hash functions :

<https://www.youtube.com/watch?v=cczlpiiu42M>

### 1. MD5 :

It will convert the plain text into hexadecimal text of fixed length. It always creates a unique hash for the plain text.

## Tool we can use for hash cracking :

Hashcat is the world's fastest and most advanced password recovery tool. It is the fastest hash recovery tool which converts the wordlist into the hashes and then

matches those hashes with the specific hash we want to recover. It is pre-installed in kali linux OS.

Instead of using standard CPU cores, it will use GPU or Graphic card cores.

### Commands :

```
#hashcat -m 0 -a 3 <hashfile in txt> <dictionary|wordlist>
```

```
#hashcat -m 0 -a 3 /root/Desktop/hash.txt /usr/share/wordlists/rockyou.txt
```

```
#hashcat -m 0 -a 3 /root/Desktop/hash.txt /usr/share/wordlists/rockyou.txt --  
force
```

Here,

hashcat is the tool for password recovery

-m : hash type

0 : MD5

-a : attack mode

3 : Brute force attack

hash.txt : file containing hashes to be recovered

rockyou.txt : for brute forcing and comparing

--force : to start forcefully